

SuperBuzz PPC Analytics

Appendix: Global Filter System

Platform Filtering & Customer Segmentation

Handover Documentation

May 2026

Appendix: Global Filter System

Supplement to the main project handover document. Covers the two filter dimensions that must run across every page, every API call, and every analysis engine in the app.

Why This Matters

This app exists to help DTC brands make better spend decisions. A blended "all channels, all customers" view is the default starting point -- but it hides the signal. The real decisions happen when a CMO asks:

- "Is Google Ads actually acquiring new customers, or just recapturing existing ones?"
- "What does Meta look like if I strip out returning customers?"
- "Show me non-brand performance on Google only -- new customers only."

Without these filters, the app is a reporting tool. With them, it becomes a decision system.

The Two Filter Dimensions

Filter 1: Platform

Controls *which* advertising channel or data source the user is looking at.

All Platforms ▾	
All Platforms	← default, aggregated view
Google Ads	← Search + Shopping + PMax
Meta Ads	← Facebook + Instagram
TikTok Ads	← placeholder / future
Google Search Console	← organic search data
Shopify	← order-level data

When a specific platform is selected, an optional sub-channel dropdown appears:

Platform	Sub-channels
Google Ads	Google Search, Google Shopping, Performance Max
Meta Ads	Prospecting / LAL, Retargeting
Shopify	Paid (UTM-attributed), Organic, Direct, Email, Referral
TikTok Ads	(none yet)
GSC	(none)

"All Platforms" aggregates everything. This is the default. Every number shown is the sum across all connected sources.

Filter 2: Customer Segment

Controls *which customer type* the metrics represent.

[All Customers] [New] [Existing]

Segment	What it means
All Customers	Total metrics -- new + returning combined. Default view.
New Customers	Only first-time buyers. Revenue, conversions, CPA, ROAS -- all recalculated for new customers only.
Existing Customers	Only returning/repeat buyers. Same recalculation, opposite segment.

How Metrics Change Per Segment

This is the critical part. When the user switches segment, every derived metric on every page recalculates:

"All Customers" (default)

Metric	Calculation
Revenue	Total revenue from all orders
Conversions	Total conversion count (new + returning)
CPA	Total cost / total conversions
ROAS	Total revenue / total cost
MER	Total revenue (all sources) / total ad spend
New Customer %	new_customers / total_conversions
CVR	Total conversions / total clicks

"New Customers" selected

Metric	Calculation
Revenue	Estimated new customer revenue (see estimation method below)
Conversions	<code>new_customers</code> field from DailyMetrics
CPA	Total cost / new_customers
ROAS	New customer revenue / total cost
MER	Total new customer revenue / total ad spend
New Customer %	100% (by definition)
CVR	new_customers / total clicks

"Existing Customers" selected

Metric	Calculation
Revenue	Estimated returning customer revenue
Conversions	<code>returning_customers</code> field from DailyMetrics
CPA	Total cost / <code>returning_customers</code>
ROAS	Returning customer revenue / total cost
MER	Total returning customer revenue / total ad spend
New Customer %	0% (by definition)
CVR	<code>returning_customers</code> / total clicks

Revenue Estimation Method

The data model stores `revenue` as a single total per campaign per day. It does not split revenue by customer type at the daily metric level.

Recommended approach (ratio-based estimation):

```
new_customer_revenue = revenue * (new_customers / (new_customers + returning_customers))
returning_customer_revenue = revenue - new_customer_revenue
```

This is an approximation. It assumes new and returning customers have similar AOV within a campaign on a given day. For most DTC brands this is directionally accurate. The UI should show a small disclaimer: *"Revenue split estimated from customer count ratios."*

More accurate alternative (when Shopify is connected):

The `ShopifyOrder` table has a `new_customer` boolean on each order. When Shopify data is available, revenue can be split precisely at the order level and aggregated up. This should be preferred when the data exists.

How Both Filters Combine

The two filters are independent and stack. Every query to the backend should pass both:

```
GET /api/v1/metrics/kpi-summary?platform=google_ads&customer_segment=new&start=2026-01-01&end=2026-03-31
```

This returns: KPI summary for Google Ads, new customers only, Q1 2026.

Example Combinations

Platform	Segment	What the user sees
All	All	Full blended view (default)
All	New	New customer metrics across all channels
Google Ads	All	All Google Ads metrics, both customer types
Google Ads	New	Google Ads new customer acquisition only
Meta Ads	Existing	Meta retargeting / re-engagement performance
Google Ads > Search	New	Google Search new customer performance only

The Question Every Number Must Answer

*"This metric is for **[platform]** **[segment]** over **[date range]**."*

If a user cannot immediately tell which platform and which customer segment a number represents, the filter implementation is incomplete.

Page-by-Page Filter Impact

Overview

Element	Platform filter effect	Customer segment effect
KPI cards (Spend, Revenue, ROAS, MER, CPA, New %)	Scoped to selected platform's campaigns	Recalculated for selected segment
Revenue & Spend trend chart	Only selected platform's daily data	Revenue line reflects segment's share
Brand vs Non-brand donut	Split within selected platform only	Segment affects revenue attribution
Top Insights	Only insights relevant to selected platform	Insights reference selected segment
Top Recommendations	Actions relevant to selected platform	Recommendations adjust for segment
Health Indicators	Brand dependence calculated per-platform	Segment changes what "healthy" looks like

Brand vs Non-brand

Element	Platform filter effect	Customer segment effect
Spend / revenue split	Brand vs nonbrand within selected platform	Revenue split recalculated per segment
Performance comparison table	ROAS, CPA, CVR per platform	All metrics recalculated for segment
Search term classification	Terms from selected platform only	No direct effect (terms are pre-customer)
Trend charts	Time series scoped to platform	Revenue/conversion lines per segment

Key insight this enables: "Non-brand on Google drives 56% of new customers but only 23% of returning customer revenue" -- this is invisible without the segment filter.

Incrementality

Element	Platform filter effect	Customer segment effect
Observational comparison	Brand vs nonbrand within platform	Metrics per segment
Cannibalization estimate	Paid vs organic for selected platform	New customer cannibalization vs total
Spend-response curves	Curves fitted per platform	Curves fitted per segment
Marginal ROAS	Calculated per platform	Different by segment (nonbrand mROAS for new customers vs existing)
Experiment results	Experiments targeting selected platform	Lift broken down by segment

Key insight this enables: "Brand spend cannibalization is 72% overall, but only 15% for new customers -- brand ads do help acquire first-time buyers, just not returning ones."

Causation Lab

Element	Platform filter effect	Customer segment effect
Correlation matrix	Correlations computed from platform-specific data	Computed using segment-specific outcome variables
Variable pair analysis	All 5 causal checks scoped to platform	"Does nonbrand spend cause new customer growth?" vs "Does it cause total revenue?"
Causal confidence scores	Recalculated per platform (different data, different verdict)	Recalculated per segment

Key insight this enables: "Google nonbrand spend has HIGH causal confidence for new customer growth, but only MEDIUM for total revenue" -- the causal story changes by segment.

Pattern Memory

Element	Platform filter effect	Customer segment effect
Current state panel	Reflects selected platform's current spend/traffic	No direct effect on state definition
Similar states	Historical matching uses platform-specific metrics	Outcome summaries show segment averages
Outcome summary	Avg revenue, CVR, marginal return for platform	"In similar conditions, new customer CVR was X%"
Pattern insights	Platform-specific: "Google Search performed Y in this context"	"New customer acquisition was Z in similar conditions"

Recommendations

Element	Platform filter effect	Customer segment effect
Recommendation cards	Platform-specific actions: "Reduce Google brand spend" vs "Increase Meta prospecting"	Segment-aware rationale: "...because new customer CPA on Google brand is 3x higher than Meta"
Cross-platform actions	Only when "All Platforms" is selected	"Shift budget from Google brand to Meta prospecting for better new customer efficiency"
Confidence scores	Different per platform (data quality varies)	Different per segment (smaller sample = lower confidence)

Experiments

Element	Platform filter effect	Customer segment effect
Experiment list	Filtered by which platform the experiment targets	No filter (show all experiments)
Results	Results scoped to experiment's platform	Lift broken down: "Total lift: 12%. New customer lift: 18%. Existing customer lift: 4%."
Design form	Platform selection as part of experiment setup	Optional segment targeting in config

Scenario Simulator

Element	Platform filter effect	Customer segment effect
Input sliders	"Change Google Ads brand spend by X%" (platform-specific)	No input change, but output changes
Projected results	Projections based on platform-specific response curves	Shows segment breakdown: "New customers: +15%. Existing: -2%."
Historical analogues	Matched from platform-specific history	Analogue outcomes split by segment
Warnings	Platform-specific: "Limited Google Shopping history for this scenario"	Segment-specific: "Insufficient new customer data for high confidence"

Admin

Element	Platform filter effect	Customer segment effect
Brand keywords	Cross-platform (no filter)	No effect
Classification rules	Cross-platform (no filter)	No effect
Coverage stats	Filterable by platform: "83% classified for Google, 67% for Meta"	No effect
Contextual data	Cross-platform (no filter)	No effect

What Exists Today vs What Needs Building

Currently Built

Capability	Status
Date range filter (7D, 30D, 90D, YTD, 1Y, All)	Working on Overview and Brand/Non-brand pages
<code>Campaign.channel</code> field in data model	Exists
<code>Campaign.data_source_id</code> FK in data model	Exists
<code>DailyMetrics.new_customers</code> field	Exists
<code>DailyMetrics.returns_customers</code> field	Exists
<code>ShopifyOrder.new_customer</code> boolean	Exists
Backend <code>channel</code> param on campaigns endpoint	Exists

Not Yet Built

Capability	What's needed
Platform filter UI component	New <code>PlatformFilter</code> dropdown component
Customer segment toggle UI	New <code>CustomerSegmentToggle</code> three-button component
Global filter context	New React context wrapping the app, persisting filter state across pages
Backend platform param on all metric endpoints	Add <code>platform</code> query param to ~15 endpoints
Backend segment param on all metric endpoints	Add <code>customer_segment</code> query param to ~15 endpoints
Revenue estimation logic	Backend service function to split revenue by customer type
Engine filter pass-through	All 7 engines need platform + segment parameters
Filter bar in page header	Update <code>Header</code> and <code>PageLayout</code> components
Empty states for filtered views	Graceful handling when filter combo returns no data

API Contract

Every metric endpoint should accept these query parameters:

```
?platform=all                # default
?platform=google_ads
?platform=meta_ads
?platform=tiktok_ads
?platform=gsc
?platform=shopify

?channel=google_search       # optional, only when platform is set
?channel=google_shopping
?channel=google_pmax
?channel=meta_prospecting
?channel=meta_retargeting

?customer_segment=all        # default
?customer_segment=new
?customer_segment=existing

?start=2026-01-01           # existing date range
?end=2026-03-31
```

Backend implementation: filter by joining `DailyMetrics` to `Campaign` where `Campaign.channel` or `DataSource.source_type` matches the platform param. For customer segment, adjust the returned metrics using the ratio-based revenue estimation or Shopify order-level data.

Edge Cases and Warnings

Small Sample Sizes

When filtering to a specific platform + new customers only, sample sizes shrink. The UI should: - Show sample size alongside metrics - Reduce confidence levels automatically when $n < 30$ days - Display a warning: "Based on limited data (X days). Interpret with caution."

Platforms with No Customer Split

Some data sources (e.g., Google Search Console) don't have customer-level data. When customer segment filter is active: - Show metrics that are available (traffic, impressions) - Grey out or hide metrics that require customer data (CPA, ROAS per segment) - Display: "Customer segmentation not available for this data source"

Cross-Platform Revenue Attribution

When viewing "All Platforms" with a customer segment filter, revenue attribution across platforms may overlap (same customer, multiple touchpoints). The UI should note: "Revenue is attributed per-platform and may overlap in the All view."

Zero-Division Protection

When a segment has zero customers (e.g., no new customers on a slow day), CPA and CVR become undefined. Show "--" or "N/A" rather than Infinity or NaN.

Summary for Developers Picking This Up

1. **Two filters, always present:** Platform (dropdown) and Customer Segment (toggle). Both appear on every page.
2. **Three things always active:** Platform + Segment + Date Range. Every API call passes all three.
3. **Global state:** Use React context. Filters persist across page navigation.
4. **Backend changes:** Add `platform` and `customer_segment` params to every endpoint that returns metrics.
5. **Revenue split:** Use ratio-based estimation from `new_customers` / `returning_customers` counts. Upgrade to Shopify order-level data when available.
6. **Engines:** Every analysis engine (classification, incrementality, causation, patterns, insights, recommendations, simulator) must accept and respect both filter dimensions.
7. **Edge cases:** Handle small samples, missing data, zero-division, and cross-platform overlap gracefully.
8. **Reference:** See `docs/filtering-specification.md` for the full technical spec with implementation phases.